

Problem Set 1

Applied Stats/Quant Methods 1

Due: October 8, 2026

Instructions

- Please show your work! You may lose points by simply writing in the answer. If the problem requires you to execute commands in R, please include the code you used to get your answers. Please also include the .R file that contains your code. If you are not sure if work needs to be shown for a particular problem, please ask.
- Your homework should be submitted electronically on GitHub.
- This problem set is due before 23:59 on Thursday October 8, 2026. No late assignments will be accepted.

Question 1: Education

A school counselor was curious about the average of IQ of the students in her school and took a random sample of 25 students' IQ scores. The following is the data set:

```
1 y <- c(105, 69, 86, 100, 82, 111, 104, 110, 87, 108, 87, 90, 94, 113, 112, 98,  
      80, 97, 95, 111, 114, 89, 95, 126, 98)
```

1. Find a 90% confidence interval for the average student IQ in the school (As a hint, you first need the mean and SD to create a CI):

```
1 sample_mean <- mean(y) #the mean is 98.44  
2 sample_sd <- sd(y) #the standard deviation is 13.09287  
3 sample_size <- length(y) #the length is 25  
4  
5 conf_level <- 0.90
```

```

6 alpha <- 1 - conf_level #confidence interval: 1 - 0.90 = 0.10
7 df <- sample_size - 1 #the degrees of freedom is 24
8
9 t_critical <- qt(1 - alpha / 2, df) # The sample size is 25 and the
    population standard deviation is unknown,so a t-test is appropriate.
10 margin_of_error <- t_critical * sample_sd / sqrt(sample_size)
11
12 confidence_interval <- c(
13   lower = sample_mean - margin_of_error ,
14   upper = sample_mean + margin_of_error
15 )
16
17 confidence_interval
18 t.test(y, conf.level = 0.90) # Check using R's built-in function

```

The 90% confidence interval is 93.96 to 102.92, thus we are 90% confident that the population mean IQ of students at the school is between the lower and upper limits of the confidence interval.

2. Next, the school counselor was curious whether the average student IQ in her school is higher than the average IQ score (100) among all the schools in the country.

Using the same sample, conduct the appropriate hypothesis test with $\alpha = 0.05$.

```

1 # IQ needs to be greater, thus alternative needs to be greater.
2 iq_test <- t.test(y, mu = 100, alternative = "greater", conf.level =
    0.95)
3 iq_test
4 ifelse(iq_test$p.value < 0.05,
5   "We reject H0",
6   "We fail to reject H0")
7 #Decision: If the p-value < 0.05, reject H0; otherwise, fail to reject H0
    . The p-value = 0.7215 thus at the 5% significance level, there is
    insufficient evidence to conclude that the average student IQ is
    greater than 100. Fail to reject H0

```

Question 2: Political Economy

Researchers are curious about what affects the amount of money communities spend on addressing homelessness. The following variables constitute our data set about social welfare expenditures in the USA.

State	50 states in US
Y	per capita expenditure on shelters/housing assistance in state
X1	per capita personal income in state
X2	Number of residents per 100,000 that are "financially insecure" in state
X3	Number of people per thousand residing in urban areas in state
Region	1=Northeast, 2= North Central, 3= South, 4=West

Explore the `expenditure` data set and import data into R.

- Please plot the relationships among Y , $X1$, $X2$, and $X3$? What are the correlations among them (you just need to describe the graph and the relationships among them)?

```
1 pairs(
2   expenditure[, c("Y", "X1", "X2", "X3")]
3 )
4
5 expenditure_cor <- cor(expenditure[, c("Y", "X1", "X2", "X3")])
6 expenditure_cor
7
8 expenditure_cor_no_diag <- expenditure_cor
9 diag(expenditure_cor_no_diag) <- NA
10 expenditure_cor_no_diag
11
12 expenditure_relationship <- matrix(
13   ifelse(
14     expenditure_cor_no_diag >= 0.7, "Strong Positive",
15     ifelse(
16       expenditure_cor_no_diag >= 0.3, "Moderate Positive",
17       ifelse(
18         expenditure_cor_no_diag > 0, "Weak Positive",
19         ifelse(
20           expenditure_cor_no_diag <= -0.7, "Strong Negative",
21           ifelse(
22             expenditure_cor_no_diag <= -0.3, "Moderate Negative",
23             ifelse(
24               expenditure_cor_no_diag < 0, "Weak Negative",
25               "No Relationship"
26             )
27           )
28         )
29       )
30     )
31   )
```

```

28     )
29   )
30 )
31 ),
32 nrow = nrow(expenditure_cor_no_diag),
33 dimnames = dimnames(expenditure_cor_no_diag)
34 )
35 # (Statistics Resources: Correlation, 2026)
36 expenditure_relationship
37 # The scatterplot matrix should show upward-sloping patterns between all
    pairs of variables, consistent with their positive correlations.
38 # Statistics Resources: Correlation. (2026, Sep 29). Retrieved from
    National University: https://resources.nu.edu/statsresources/
    correlation

```

- Please plot the relationship between Y and *Region*? On average, which region has the highest per capita expenditure on housing assistance?

```

1  expenditure$Region,
2  levels = c(1, 2, 3, 4),
3  labels = c("Northeast", "North Central", "South", "West")
4 )
5
6 boxplot(
7   Y ~ RegionName,
8   data = expenditure,
9   xlab = "Region",
10  ylab = "Per Capita Expenditure on Housing Assistance",
11  main = "Per Capita Expenditure by Region"
12 )
13
14 regional_means <- aggregate(Y ~ RegionName, data = expenditure, mean)
15 regional_means
16 regional_means[which.max(regional_means$Y), ]
17 # The West (Region 4) has the highest average expenditure, with a average
    mean of 88.31 per capita. The North Central region (Region 2) has the
    second-highest average at 83.92, followed by the Northeast (Region 1)
    at 79.44. The South (Region 3) has the lowest average expenditure at
    69.19 per capita.

```

- Please plot the relationship between Y and $X1$? Describe this graph and the relationship. Reproduce the above graph including one more variable *Region* and display different regions with different types of symbols and colors.

```

1   xlab = "Per Capita Personal Income (X1)",
2   ylab = "Per Capita Expenditure on Housing Assistance (Y)",
3   main = "Relationship Between Y and X1",
4   pch = 19,
5   col = "steelblue")

```

```

6
7 abline(lm(Y ~ X1, data = expenditure),
8         col = "red",
9         lwd = 2)
10
11 model <- lm(Y ~ X1, data = expenditure)
12 summary(model)$r.squared
13 coef(model)
14
15 cor_y_x1 <- cor(expenditure$Y, expenditure$X1)
16 round(cor_y_x1, 3)
17 # The scatterplot shows a moderate positive relationship between X1 and Y
   :as per-capita personal income (X1) increases, per-capita expenditure
   on housing assistance (Y) generally increases. The correlation between
   the two variables is 0.532, indicating a moderate positive linear
   association.
18 # The abline's formula is  $Y = 32.546 + 0.02458X_1$ . Approximately 28.27%
   of the variation in Y is explained by X1 using this linear regression
   model.
19
20 plot(expenditure$X1, expenditure$Y,
21       xlab = "Per Capita Personal Income (X1)",
22       ylab = "Per Capita Expenditure on Housing Assistance (Y)",
23       main = "Relationship Between Y and X1 by Region",
24       pch = c(16, 17, 15, 18)[expenditure$Region],
25       col = c("red", "blue", "green", "purple")[expenditure$Region])
26
27 legend("topright",
28       inset = c(0.725, 0),
29       legend = c("Northeast", "North Central", "South", "West"),
30       pch = c(16, 17, 15, 18),
31       col = c("red", "blue", "green", "purple"),
32       title = "Region",
33       cex = 0.5,
34       xpd = TRUE)
35
36 par(xpd = NA)

```